

Exam Date & Time: 29-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

PROGRAMMING IN C plus plus [NPT-4080]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Using the concept of function overloading write a program in C++ to find the maximum of three integer numbers. (5)
 - A)
 - B) Develop a C++ Program to Reverse a number. (3)
 - C) Explain the destructors in C++ with programming examples of each. (2)
- 2) Create a class called clock with data members as hour, minute and member functions readtime(), showtime(). Write a C++ program to input two clock objects and add using operator overloading. (5)
 - A)
 - B) Define the importance of friend function with an example program. (3)
 - C) Illustrate enumerated data type, constant and volatile in C++. (2)
- 3) Explain the concept of virtual base class with programming example. (5)
 - A)
 - B) Explain the concept of inheritance in C++ with a programming example. (3)
 - C) Explain the concept of protected base class inheritance in C++. (2)
- 4) Explain the different types of polymorphism in C++ with Programming example. (5)
 - A)
 - B) Analyze the importance of run time and compile time polymorphism in C++ with a programming example. (3)
 - C) Explain the rules of virtual functions in C++. (2)
- 5) Explain the concept of stream class hierarchy in I/O systems. (5)

A)

B) Explain the concept of formatted I/O in C++ with programming examples. (5)

-----End-----